MIL-C-40004B
15 April 1969

SUPERSEDING
MIL-C-40004A
14 May 1964

MILITARY SPECIFICATION

CLOTH, KNITTED, COTTON, SIMPLEX

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers two types of simplex knitted cotton cloth.

1.2 Classification.- The cloth shall be of the following types and classes as specified (see 6.2):

Type I - 9.5 oz. per sq. yd.
Type II - 7.5 oz. per sq. yd.
Class 1 - White, Shade No. 3028
Class 2 - Gray, Shade No. 1164
Class 3 - Gray, Shade No. 1163
Class 4 - Seal Brown, Shade No. 105
Class 5 - Gray Beige, Shade No. 270
Class 6 - Black, Shade No. 3226

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

* PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

FSC 8305

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STANDARDS

FEDERAL

Fed. Test Method Std. No. 191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D. C. 20580.)

3. REQUIREMENTS

3.1 Standard sample.- The dyed and finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard is referenced (see 6.3).

* 3.2 First article.- When specified (see 6.2), before production is commenced, 3-1/2 yards, full width, of the finished cloth shall be submitted or made ready for the contracting officer or his authorized representative for inspection as specified in 4.2. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured by the supplier in the same facilities to be used for the manufacture of the production cloth.

3.3 Material.-

* 3.3.1 Yarn.- The yarn shall be made from cotton which has been carded, combed and spun into singles yarn.

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* 3.4 Color.- The color shall be as specified (see 6.2). When white (class 1) is specified, the cloth shall be bleached and shall be supplemented with an optical brightener to enhance the appearance of cloth. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.5. When a standard sample is not available, the dyed and finished cloth shall show no more than a slight trace of labile sulfur as defined in the test method specified in 4.5.

3.4.1 Matching.- The color of the finished cloth shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.4.2 Colorfastness.- The dyed cloth shall show fastness to laundering, perspiration, and crocking equal to or better than the standard sample when tested as specified in 4.5. When no standard sample is established, the dyed cloth shall show "good" fastness to laundering, perspiration and crocking when tested as specified in 4.5, except that class 6 cloth shall show "fair" fastness to crocking.

3.5 Physical requirements.- The dyed and finished cloth shall conform to the requirements specified in table I when tested as specified in 4.5.

TABLE I.- Physical requirements

Type	Weight per sq/yd (min)	Texture per in. (min)		Bursting strength (min)	Stretch, percent Length (Wales)		Width (Courses)
	Ounces	Wales	Courses	Pounds	Max	Min	Max
I	9.5	60	42	130	15	40	60
II	7.5	66	40	120	20	60	80

3.5.1 Width.- Unless otherwise specified (see 6.2), the overall minimum width shall be 36 inches.

3.5.2 Knit.- The knit shall be a 48-line Atlas pattern (see 6.4). The cloth shall be knit with nonravelling edges (see 6.5).

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3.6 Finish.- The cloth shall be sueded on both sides and have a suppleness equal to or better than the standard sample.

3.6.1 Nonfibrous materials.- The starch and protein content including chloroform-soluble and water-soluble material of the finished cloth shall not exceed 5.0 percent when tested as specified in 4.5.

3.7 pH.- The pH value of the water extract of the finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.5.

* 3.8 Dimensional stability.- There shall be no more than 10.0 percent shrinkage in the direction of the length (wale direction) or 3.0 percent in the width (course direction) when tested as specified in 4.5. The preshrinkage process used shall not be identified by name or trademark either on the cloth, ticket or package.

* 3.9 Length and put-up.- Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 35 yards. Each length shall be put-up on full width rolls as specified in PPP-P-1134.

3.10 Fiber identification.- Each roll shall be labeled, ticketed or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.11 Workmanship.- The finished cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Certificate of compliance.- Where certificates of compliance are submitted the Government reserves the right to check test such items to determine the validity of the certification.

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* 4.2 First article inspection.- When required, the preproduction sample submitted in accordance with 3.2 shall be visually examined for appearance, color and finish. The sample shall be tested for physical and chemical properties in accordance with 4.5, to the extent applicable.

* 4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 Examination of the end item.- Examination of the end item shall be in accordance with 4.3.2.1 through 4.3.3.2.

4.3.2.1 Yard-by-yard examination.- The required yardage of each roll shall be examined and visual defects classified as listed below. All defects found shall be counted regardless of their proximity one to another, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each lengthwise yard or fraction thereof in which it occurs. The sample unit for this examination shall be one linear yard. The sample size shall be in accordance with the inspection level II of MIL-STD-105. The AQL shall be 1.5 major and 6.5 total defects (major and minor combined) per 100 units (yards). The lot size shall be expressed in units of one linear yard each. An approximate equal number of yards shall be examined from each roll selected in accordance with table II.

Defects	Classification Major	Minor
Hole, cut, tear, any	X	
Run or dropped stitch	X	
Tuck stitch, pulled stitch or needle streak		X
Mend	X	
Abrasion mark 1/	X	
Thin or uneven cloth 1/	X	
Course or fine yarn 1/	X	
Tight twist yarn		X
Slubs or gouts, more than twice the size of normal yarn		X
Specks 1/		X

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Defects (cont'd)	Classification Major	Minor
Knots extending above the surface of cloth 1/		X
Kinks or untrimmed ends		X
Crease or wrinkle, (embedded)		X
Stretch (stitch tension) knitting tension too tight or too loose	X	
Shade bar 1/		X
Spot or stain	X	

1/ Clearly visible at normal inspection distance (3 feet).

4.3.2.2 Overall examination.- Each defect listed shall be counted not more than once in each roll examined. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the acceptable number (maximum number of defects acceptable) shall be as shown in table II.

Defects

Preshrinkage process identified by name or trademark on the cloth or ticket.
Degree and character of finish not equal to standard sample.
Degree and character of sueding not equal to standard sample.
Objectionable odor.
Overall uncleanness.
Uneven knitting throughout the roll.
Off shade, shaded end to end, side to side, side to center or throughout the roll.
Mottled, streaky or cloudy in appearance.
Edges frayed, ravelled, curled, rolled, folded, scalloped or uneven continuously or intermittently throughout the roll.
Not labeled in accordance with Textile Fiber Products Identification Act.
Overall width less than specified.

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* TABLE II.-

Lot size in yards	Sample size in rolls	Maximum number of defects acceptable in sample
Up to 1200 1/	3	0
1201 up to and including 3200	5	0
3201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,001 and over	32	2

1/ If lot is fewer than three rolls, each roll in the lot shall be examined.

4.3.3 Examination for length.-

4.3.3.1 Individual rolls.- During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The maximum number of defects acceptable in the sample shall be in accordance with table II.

4.3.3.2 Total yardage in sample.- The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

* 4.4 Examination of preparation for delivery requirements.- An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking requirements of section 5 of this specification are complied with.

4.5 Testing of the end item.- The methods of testing specified in Fed. Std. No. 191, wherever applicable, and as listed in table III shall be followed. The physical values specified in section 3, except where otherwise specified, apply to the results of the determinations made on the sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 3-1/2 yards, full width of the finished cloth. The lot size shall be expressed in units of one yard each. The lot shall be unacceptable if one or more units fail to meet any test requirement specified. The sample size (number of sample units) shall be in accordance with the following:

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Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

* TABLE III.- Test methods

Characteristic	Requirement paragraph	Test method
Material identification (cotton)	3.3.1	1200 1/
Presence of labile sulfur	3.4	2020
Optical brightener (class 1)	3.4	1/
Colorfastness to:		
Laundering	3.4.2	5614 2/
Perspiration	3.4.2	5680
Crocking	3.4.2	5651
Weight	3.5	5041
Texture (wales and courses)	3.5	5070
Bursting strength	3.5	5120
Stretch (percent)	3.5	4.5.1
Knit	3.5.2	1/
Nonfibrous material	3.6.1	2611
Acidity (pH)	3.7	2811
Dimensional stability	3.8	5550

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ For classes 2 and 3 cloths, the temperature of the water during the test shall be 71° ± 1°C.

4.5.1 Test for stretch.-

4.5.1.1 Test specimen.- The test specimen shall be a rectangle 4 inches by not less than 5 inches. The long dimension shall be parallel to the length (wales) of the cloth for length tests and parallel to the width (courses) for width tests. The test specimen shall be taken no nearer the selvage than 1/10 the width of the cloth. Two parallel lines, 3 inches apart shall be drawn at right angles to the long dimension of the specimen. In addition, a line shall be drawn parallel to the long dimension of the specimen and 1 inch in from either edge for properly aligning the specimen in the jaws.

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4.5.1.2 Test apparatus.- The test device (figure 1) shall be an apparatus wherein the specimen is held between two clamps. The design of the upper clamp shall be such that one gripping surface or jaw shall be an integral part of the rigid frame of the clamp. The face of the movable jaw shall measure 1 inch by 1 inch and the face of the fixed jaw shall measure 1 inch by 1-1/2 or more inches with the long dimension perpendicular to the direction of application of the load. The lower clamp shall have similar jaws and, when weight is attached, shall weigh a total of 6 pounds. The jaws shall have smooth gripping surfaces, sufficiently flat and parallel to prevent slipping of the specimen during the test.

4.5.1.3 Procedure.- The specimen shall be placed symmetrically in the clamps of the apparatus with the long dimension parallel and the short dimension at right angles to the direction of the application of the load. Using the specified guidelines, the specimen shall be aligned with one outside edge of the front jaws of each clamp. The proper weight shall be added to insure a total of 6 pounds (including the weight of the lower jaws). The load shall be applied for 15 seconds and the distance between jaws measured by a pair of dividers or other suitable instrument. The elongation at the required load shall be expressed as the percent increase in the length of the specimen between the jaws. The percent stretch shall be calculated as follows:

$$\text{Percent stretch} = \frac{\text{Stretch dimension - Original 3-inch dimension}}{\text{Original dimension (3 inches)}} \times 100$$

4.5.1.4 Report.- One specimen each from the length (wales) and width (courses) direction shall be tested from each sample unit and the results obtained reported separately to the nearest percent.

* 5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C.- The cloth, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C.- The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking.- In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

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6. NOTES

6.1 Intended use.- The cloth is intended for use in the fabrication of gloves, scarfs and lining of protective headgear, and earphones and other personal equipment.

* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and class (color) required (see 1.2 and 3.4).
(c) First article.- When preproduction sample is required (see 3.2).
(d) Width required if other than specified (see 3.5.1).
(e) Length required if other than specified (see 3.9).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample.- For access to standard sample, address the procuring activity issuing the invitation for bids.

* 6.4 The 48-line Atlas pattern is produced when the warp ends continue to move in the same direction for 24 consecutive courses after which the direction of the guide bar movement is reversed and the warp ends traverse 24 courses in the opposite direction. A faint horizontal line occurs at each half repeat where the guide bars change direction. This represents one full repeat of the pattern.

6.5 Knitting machine.- The standard sample for both type I and type II cloth was knit on a 32 gauge, two-needle bar warp knitting machine (simplex).

* 6.6 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 11

Review activities:

Army - MD
Navy - SA, MC
Air Force - 82, 85

Preparing activity:

Army - GL

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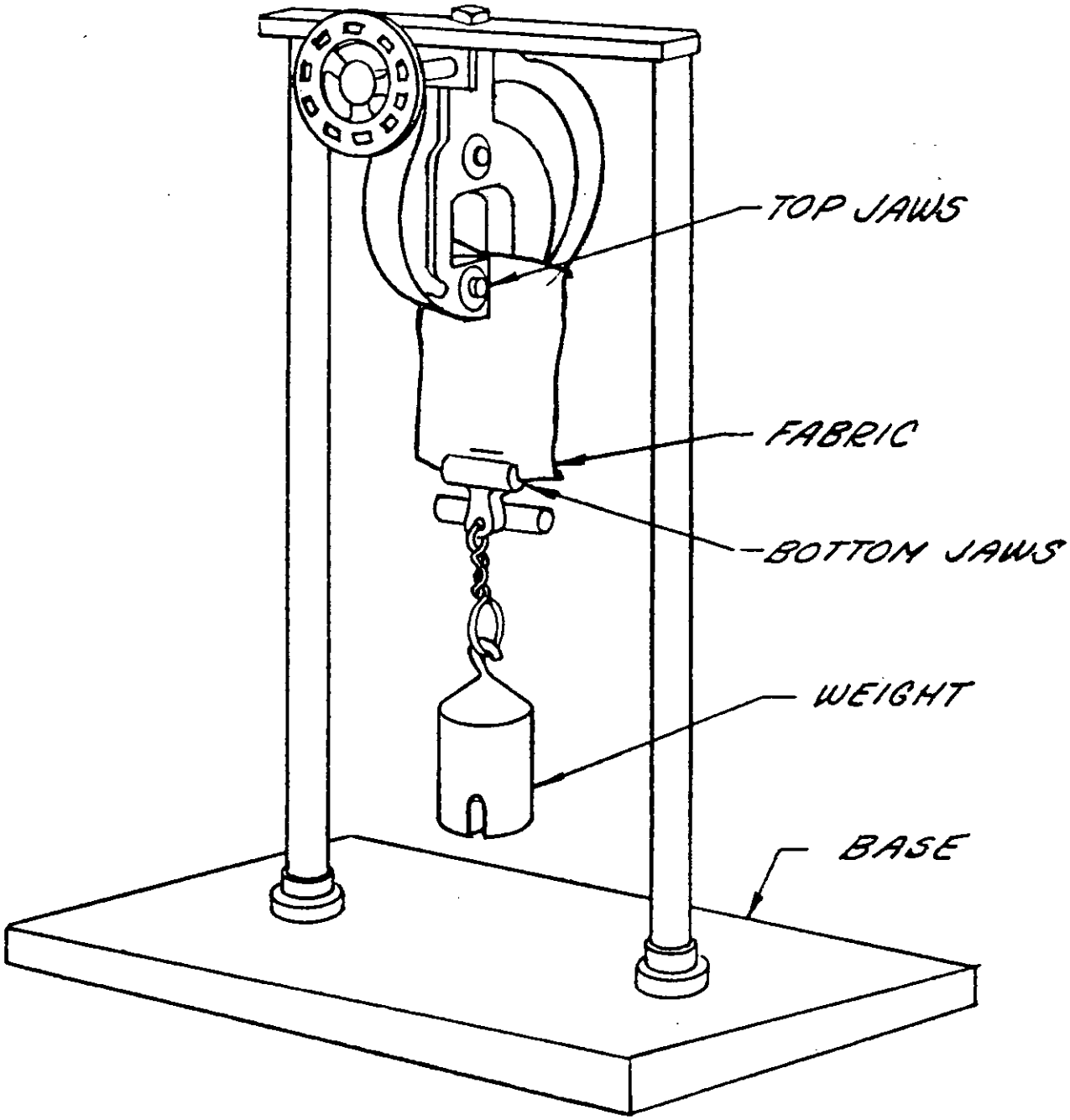


FIGURE 1. *Fabric stretch apparatus.*

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-C-40004B Cloth, Knitted, Cotton, Simplex

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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